MILITARY SPECIFICATION

PLASTIC BONDED HMX (95/5) POWDER (FOR USE IN AMMUNITION)

1. SCOPE

1.1 This specification covers an explosive designated as Plastic Bonded HMX (PBHMX) (see 6.3).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-H-45444 - HMX

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105)
MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-650 - Explosive: Sampling Inspection and Testing
MIL-STD-1168- Lot Numbering of Ammunition
MIL-STD-1171- Acceptance and Description Sheets for Propellants and Explosives
MIL-STD-1235- Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

FSC: 1376

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DRAWINGS

ORDNANCE CORPS

7548644	-	Box, Packing for High Explosives Assembly, Details, Packing and Marking
7548645	-	Carton, Packing, Reusable, Collapsible for High Explosives

(Copies of standards, specifications, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer).

2.2 Other publications.-The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM Designation: E300-70 Recommended Practice for Sampling Industrial Chemicals

(Application for copies of ASTM Standards should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia 3, Pennsylvania, 19103).

3. REQUIREMENTS

3.1 Material.-The material shall be HMX (complying with Grade II, Class B, of MIL-H-45444A) uniformly coated with a plastic binder (see 6.4) and shall be in the form of free flowing granular particles without visual signs of agglomeration. The composition shall be made by coating the HMX with plstic binder by the wet slurry method (see 6.5).

3.2 Composition.-The composition of the PBHMX shall be as specified in Table I, when determined as specified in applicable paragraphs of 4.4.1.

TABLE I

CONSTITUENT	PERCENT BY WEIGHT	APPLICABLE PARAGRAPHS
Plastic Binder	5.0 ± 1.0	4.4.1.1
HMX	95.0 ± 1.0	4.4.1.2

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3.3 Bulk density.-The bulk density of the PBHMX shall be 0.90 grams (g) per cubic centimeter (cc), min., when determined as specified in 4.4.2.

3.4 Moisture.-The moisture content of the PBHMX shall be 0.10 percent, maximum, when determined as specified in 4.4.3.

3.5 Workmanship.-The material shall be processed in a manner that will produce the high quality material necessary to meet the requirements of this specification. The material shall be free of dirt, chips, and other foreign matter.

3.6 First article testing.-This specification makes provisions for first article testing. Submission of first article quantity by the contractor shall be as specified in the contract.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL-STD-109 in order to define the terms used herein.

4.1.1 Submission of product.-At the time the completed lot of product is submitted to the Government for acceptance the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all requirements and quality assurance provisions specified in this specification.

b. Specification number and date, together with an identification and date of changes.

c. Certificates of analysis on all material used directly by the contractor when such material is controlled by Government specifications, shall be made available upon request by the contracting officer.

d. Quantity of product in the lot.

e. Date submitted.

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The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.2 First Article Inspection

4.2.1 Submission.-The contractor shall submit a first article sample consisting of 100 grams of PBHMX each from a minimum of two or three batches in accordance with instructions issued by the Contracting Officer for evaluation in accordance with paragraph 4.2.2. All samples submitted shall have been produced by the contractor using the same production processes, procedures, and equipment as will be used in fulfilling the contract. All materials, including packaging and packing, shall be obtained from the same sources of supply as will be used in regular production. The sample shall be accompanied by certificates of analysis. A first article quantity, shall also be submitted whenever there is a lapse in production for a period in excess of 90 days, or whenever a change occurs in manufacturing process, material used, drawing, specification or source of supply as to significantly affect product uniformity as determined by the Government. Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract and submit a record of this inspection with the sample. A sample containing known defects will not be submitted unless specifically authorized by the Contracting officer.

4.2.2 Inspections to be performed.-The sample will be subjected by the Government to any or all of the examinations or tests specified in 4.3 and 4.4 of this specification and any or all requirements of the applicable drawings.

4.2.3 Rejection.-If any sample fails to comply with any of the applicable requirements, the first article quantity shall be rejected. The Government reserves the right to terminate its inspection upon any failure of a sample to comply with any of the stated requirements.

4.3 Inspection Provisions

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4.3.1 Lot formation.-A lot shall consist of one or more batches of PBHMX produced by one manufacturer in accordance with the same specification or same specification revision under one continuous set of operating conditions. Each batch shall consist of that quantity of PBHMX that has been subjected to the same unit chemical or physical mixing process intended to make the final product homogeneous. The product shall be submitted for inspection in accordance with MIL-STD-105. The criteria and procedures for the assignment of lot numbers shall be in accordance with Standard MIL-STD-1168.

4.3.2 Examination.-Sampling plans and procedures for the following classifications of defects shall be in accordance with MIL-STD-105 (ABC-STD-105). Contractor's sampling plans, if used, shall be approved by the Government and shall provide as a minimum, the protection afforded the Government by the sampling plans in MIL-STD-105. Continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the procuring activity. Also, at the option of the procuring activity, AQL's and sampling plans may be applied to the individual characteristics listed, using an AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each Minor Defect.

4.3.2.1 Box or carton prior to closing (see dwg. 7548644 and 7548645).

Categories	Defects	Method of Inspection	Code No. (see 6.2)
Critical:	None defined.		
Major:	AQL 0.65 percent		
101.	Foreign matter	Visual	01001
102.	Liner pierced or torn	Visual	01002
103.	Liner improperly closed	Visual	01003
Minor:	AQL 0.65 percent		
201.	Type of liner incorrect	Visual	01004

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4.3.2.2 Sealed boxes (see dwg. 7548644).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.65 percent		
101.	Top improperly assembled	Visual/ Manual	02001
102.	Box damaged	Visual	02002
103.	Lot number misleading or unidentifiable	Visual	02003
104.	Strapping missing, broken, or loose	Visual/ Manual	02004
Minor:	AQL 1.00 percent		
201.	Strapping improperly assembled	Visual/ Manual	02005
202.	Marking misleading or unidentifiable	Visual	02006

4.3.2.3 Sealed fiberboard carton (see dwg. 7548645).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.40 percent		
101.	Assembly torn or pierced	Visual	03001
102.	Lot number misleading or unidentifiable	Visual	03002
Minor:	AQL 1.50 percent		
201.	Stitches missing or loose	Visual	03003
202.	Marking misleading or unidentifiable	Visual	03004
203.	Banding strips missing, broken or improperly applied	Visual/ Manual	03005

4.3.3 Sampling for Test.-The tests depicted in 4.4.1 through 4.4.3 shall be performed on a representative sample from each batch. A representative sample of 150 grams shall be selected from each batch in accordance with ASTM Procedure E300-70 for solids.

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4.3.3.1 Skip Batch Testing.-When the contractor has demonstrated that the constituent materials, production processes and testing procedures are uniform and within control, as evidenced by ten (10) consecutive batches passing the test requirements, the contractor may institute skip batch testing using the test frequencies prescribed below. The test frequencies represent the ratio of batches tested to batches produced.

NO. BATCHES IN LOT	TEST FREQUENCY
1 to 3	1/1
4 to 6	3/4
7 to 10	1/2
11 or more	1/3

4.3.3.1.1 Skip Batch Selection.-When a sufficient number of batches have been produced to permit selection at the prescribed test frequency, the contractor shall randomly select a sample from a batch for acceptance testing. Actual selection shall be accomplished by using the serial numbers of the candidate batches and applying a table of random numbers in selecting the serial number of the batch to be tested (see 6.6). As indicated in the table above, the test frequency is based on the contractor's predetermination of the lot size (number of batches in a production lot).

4.3.3.1.2 Skip Batch Test Provisions.-The contractor shall test the batch sample in accordance with the methods and procedures in paragraph 4.4. If the sample meets the test requirements, the contractor shall continue testing at the same test frequency. If the sample fails to meet the test requirements, the batch shall be rejected and the contractor shall return to 100% testing of each batch. The contractor may return to skip batch testing when:

a. Ten (10) consecutive batches, produced after the rejected batch, have been tested and passed the requirements, and

b. $\overline{X}$ number of consecutive batches, produced prior to the rejected batch, have been tested (in the order produced) and passed the requirements ($\overline{X}$ is the number of batches being skipped at the prevailing skip-batch frequency).

4.4 Test Methods and Procedures

4.4.1 Determination of Composition

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4.4.1.1 Plastic binder, Major defect, Defect Code No. 04001.- Transfer an accurately weighed portion of approximately 1 gram of the sample to a tared medium porosity fritted glass crucible. Extract the contents of the crucible with 4,25 ml portions of toluene saturated with HMX. Stir the contents of the crucible after the addition of each portion and allow the solvent to remain in contact with the sample for 3 minutes before applying suction. Aspirate the crucible and contents until the odor of toluene is no longer perceptible, dry in an oven maintained at 100° ± 5°C for 1 hour, cool in a desiccator and weigh. Calculate the loss in weight of the crucible and contents to percentage plastic binder as follows:

$$\text{Percent Plastic Binder} = \frac{100\ A}{B}$$

Where:

A = loss in weight, in g
B = weight of sample on dry basis, in g

4.4.1.2 HMX content, Major defect, Defect Code No. 05001.-The percent of HMX in the sample shall be calculated by subtracting from 100 the percent plastic binder obtained as specified in 4.4.1.1.

4.4.2 Bulk density, Major defect, Defect Code No. 06001.-The bulk density of the compound shall be determined in accordance with Method 201.3 of Standard MIL-STD-650.

4.4.3 Moisture, Major defect, Defect Code No. 07001.-The moisture shall be determined in accordance with Standard MIL-STD-650, Method 101.4, using methanol as the special solvent.

4.4.4 Granulation (see 6.7).-For informational purposes only, the granulation of the PBHMX shall be determined in accordance with Method 204.1 of the latest revision of Standard MIL-STD-650 using U. S. Standard Sieves No. 20,60, 80 and 100..

5. PREPARATION FOR DELIVERY

5.1 Packaging

5.1.1 Level A.-The PBHMX shall be packaged in accordance with Drawing 7548644.

5.1.2 Level B and C.-The PBHMX shall be packaged in accordance with Drawing 7548645.

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5.2 Packing

5.2.1 Level A.-Packing shall be in accordance with paragraph 5.1.1.

5.2.2 Level B and C.-Packing shall be in accordance with paragraph 5.1.2.

5.3 Marking.-Marking shall be in accordance with Drawing 7548644 or Drawing 7548645 as applicable.

6. NOTES

6.1 Ordering data.-Procurement documents should specify the following:

a. Title, number and date of this document.
b. Description sheets shall be prepared for each lot in accordance with Standard MIL-STD-1171.
c. Provisions for submission of first article samples.

6.2 Inspection code numbers.-The five digit code numbers assigned to the inspection herein are to facilitate future data collection and analysis by the Government.

6.3 Intended use.-The plastic bonded HMX (PBHMX) covered in this specification is intended to be pelletized for use as the main charge of the XM41 Grenade.

6.4 A plastic binder which has been found satisfactory in this composition is EXON 461, a proprietary product manufactured by Firestone Plastics Company, Pottstown, Pennsylvania. The material is a fluorine containing polymer which has the following typical resin physical properties:

Specific Gravity	1.70
Form	White Granular Powder
Relative Viscosity (1% in cyclohexanone)	1.50
Solubility	Ketones, ethers, esters, aromatic and chlorinated hydrocarbons

6.5 Process.-The following wet slurry method has been found satisfactory. EXON 461 lacquer (in acetone) at approximately 25°C is added to a hot water slurry of HMX at approximately 70°C, mixed, cooled, filtered and dried. The composition should not be manufactured by blending or mixing EXON 461 and HMX in the dry state.

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6.6 The tables of random numbers in MIL-HDBK-53 (Handbook 53) "Guide for Sampling Inspection" and ICC "Table of 105,000 Random Decimal Digits" have been found suitable.

6.7 Granulation.-Granulation testing shall be performed on each batch of PBHMX produced in order to obtain additional information on the particle size of the PBHMX composition. Results of the informational testing shall be submitted to Picatinny Arsenal, ATTN: SMUPA-QA-A-P, Dover, N.J. 07801.

CUSTODIAN:
ARMY-MU

PREPARING ACTIVITY:
ARMY-MU

Project Number: 1376-A-063

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DEPARTMENT OF THE ARMY

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OFFICIAL BUSINESS

Commanding Officer
Picatinny Arsenal
ATTN: SMUPA-NR
Dover, New Jersey 07801

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

MIL-P-50854 (MU)

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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